Amélie Barrette
CART 263 – *Creative Computation II*
April 20th 2026
Prof: Sabine Rosenberg

video presentation

https://vimeo.com/1184993936/762faeabdc?share=copy&fl=sv&fe=ci

final project

Final project live link:
https://nounoune666.github.io/cart263/projects/projectTwo/

For this project, I wanted to make something more creative than my first assignment, which was essentially a walkthrough of all the material covered in class up to that point. I opted to use Three.js since I am a fan of 3D. The website is meant to serve as a creative promotional tool for the open call to my project, *simUlator*. Several interactive elements link to the project's Instagram page, a Google Forms, and some other informational pages.

My vision was to incorporate avatars I made, which I think of as trans and non-binary avatars. The goal is to navigate through their rooms in a kind of dollhouse way, where the rooms are open and you can see inside them. The overall aesthetic was meant to evoke The Sims, and you'll also hear music from The Sims playing in the background. The underlying idea is to show that you can use avatars to express your true self.

implementation

Here is a chronological breakdown of what I implemented:

1. Created and placed walls, floors, and the ceiling.
2. Exported textures from Sims Studio and imported them into the scene.
3. Reworked avatars in Blender, added animations, then exported and imported them into the scene.
4. Added proper lighting.
5. Added raycasting.
6. Implemented mouse events for hovering and clicking.
7. Added links to websites.
8. Added other 3D meshes.
9. With Sabine, implemented a class for practice.

10. Added music.
11. Troubleshot and debugged.
12. librairies & techniques used

libraries used and techniques

I used *Three.js* as my main library, along with its addons: *OrbitControls*, the *GLTF Loader*, and *SkeletonUtils*. The walls, ceiling, and floor are all made of planes using basic geometry and materials in Three.js. For the camera, I used a perspective camera with a 75 degree field of view, positioned it at the center of the room, and used if statements to create borders that contain the movement. For the panning I used simple up and down arrow key controls.

For my avatars, I made them in The Sims, then used a program called Sims Ripper that exports them as DAE files, converted them to OBJ and assigned the materials in blender. In some cases I made changes to the materials in Photoshop. I then used Mixamo for the animations, reimported into Blender, and exported into the proper GLTF format. Most of the assets came from Sims Studio, another free program that gives you access to all the textures and 3D models from The Sims. I got the instagram model from Sketchfab. For the Google Forms model, I made it in Blender and the texture in Photoshop.

All the 3D objects were put into an array using push. A large part of the code resides in the *async addAndRun* function. Every object was assigned scaling, position, and sometimes rotation and material if it wasn't already attached. I used the mixer and clip functions for the animations. In the animate function, elapsed time is tracked. One of the models, Storm Noune, uses a timer and cycles through the array every 3 seconds. For avatars with animations I had to use *SkeletonUtils* and *clone* so the model would not disappear from the scene.

With the raycaster I created intersect variables for 6 objects. Along with mousemove, click, and mouseover events that live outside the *addAndRun* function, the cursor changes to a pointer when you hover over interactive objects to signal that they can be clicked, which eliminates the need for extra instructions. The objects also scale up slightly on hover. I really wanted to let the user explore and discover things for themselves. I also added a simple rotation to 3 of the objects.

To simplify the hover detection on one of the avatars, I placed an invisible box mesh over it. The audio had to be triggered inside the

mouseWasClicked block because browsers require a user interaction before audio can play. For the clickable objects that open websites, I used raycasting again and a simple *window.open* accompanied by the url did the trick.

The most complex part was figuring out the avatar switch for Nou Noune. With Sabine's help, we first tried checking if an intersect was detected on the avatar, storing it in *currentIntersectedObj*, and then using an if/else block to toggle between the two models by removing the current one from the scene, cloning the other, and reassigning the mixer and animation. This logic was eventually moved into the *Avatar* class and simplified into a single *switchModel* call. The lighting is straightforward: almost every object has its own spotlight, and there is a hemisphere light for general ambient lighting. The light helpers proved extremely useful during development. It’s important to note that I had athey lot of great help from Sabine for which I am very grateful. This was my first time using Three.js and it certainly won't be my last.

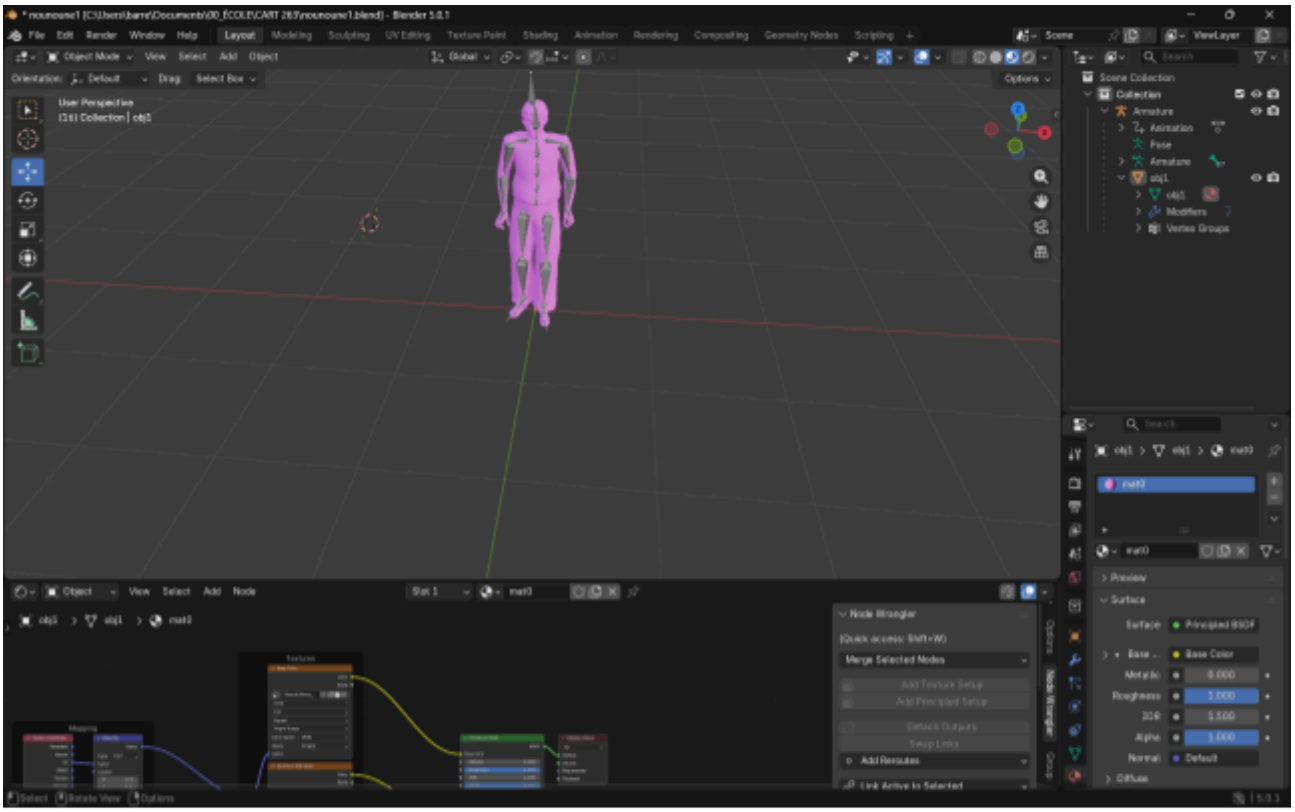
inspirations

My first inspiration comes from **Skawennati**, an artist who uses avatars from the game Second Life to create films, a practice known as *machinima*. She "makes art that addresses history, the future, and change from her perspective as an urban Kanien'kehá:ka (Mohawk) woman and as a cyberpunk avatar."* What draws me to her work is the way she uses virtual bodies / avatars as a means of self-representation and world-building.

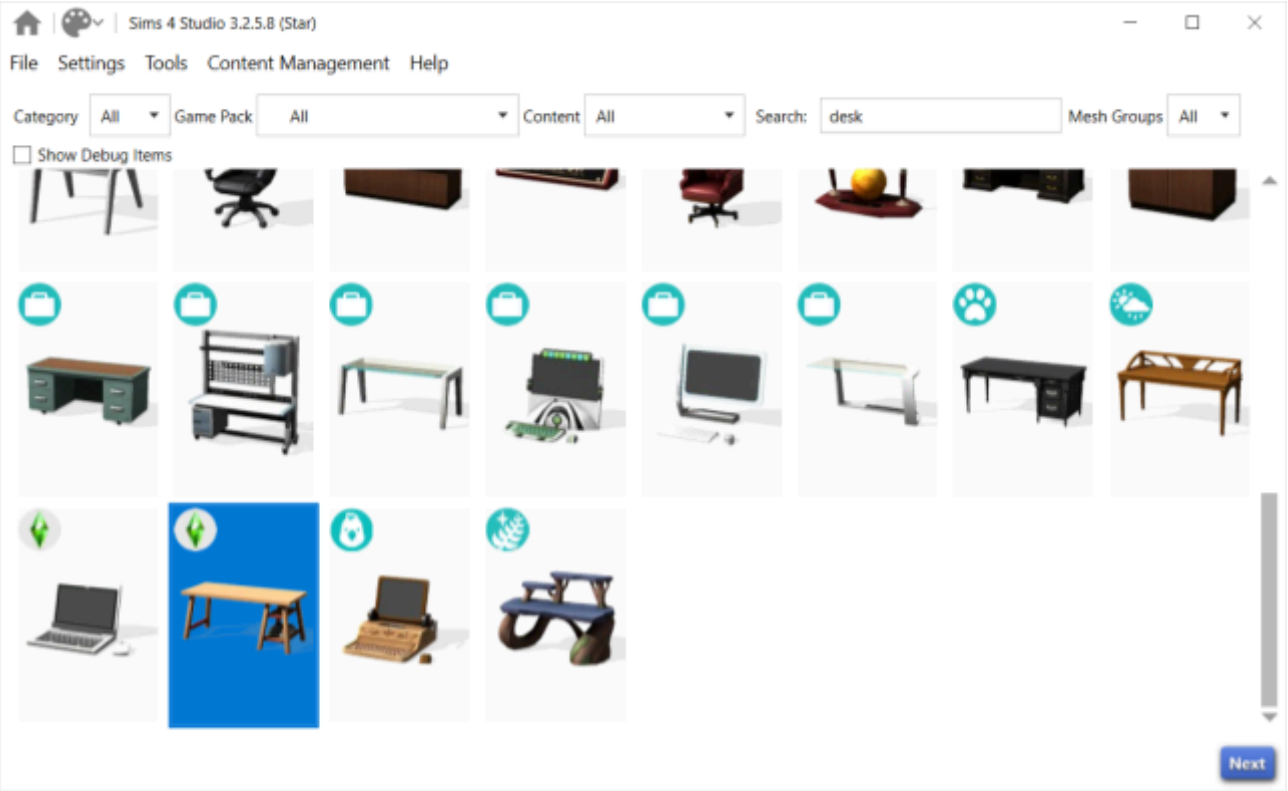
I do something similar with The Sims. The game's customizable avatars gave me a way to create characters that represent me most closely, particularly in relation to my trans identity. There is something powerful about a space where you set the rules, where the body you inhabit and the world around it are yours to define. My project *simUlator* is built on that same idea.

This website was also built with a purpose to serve as a promotional tool for the open call of ***simUlator***, a larger project I am developing in Unreal Engine. The aesthetics of the website, including the rooms, the avatars, and the overall Sims-inspired visual language, were intentionally designed to reflect the world being built in that project. In that sense, this website is both a continuation of *simUlator* and an experiment. I wanted to explore whether web-based programming and real-time 3D could be a tool for the future of the project. This assignment gave me the opportunity to test that idea and it opened possibilities for where the project could go next.

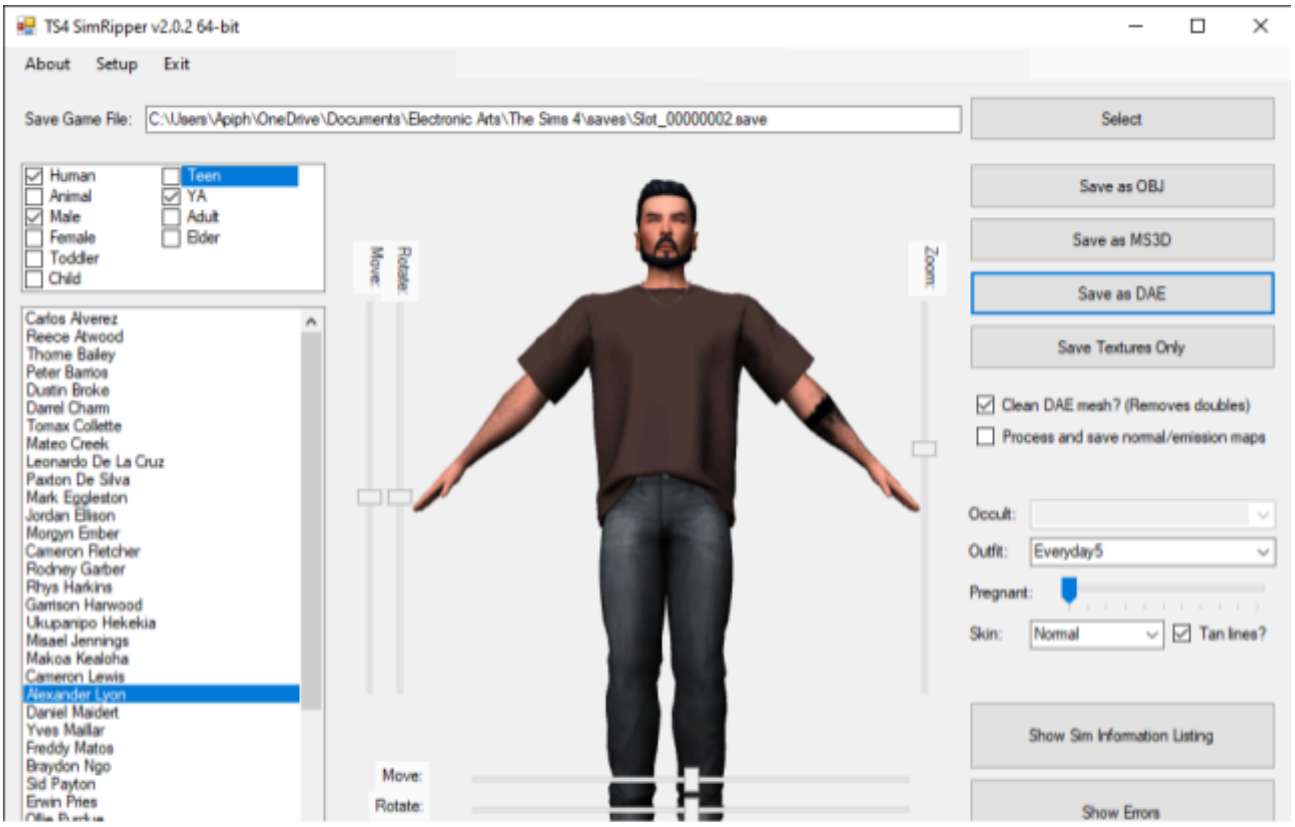
screenshots



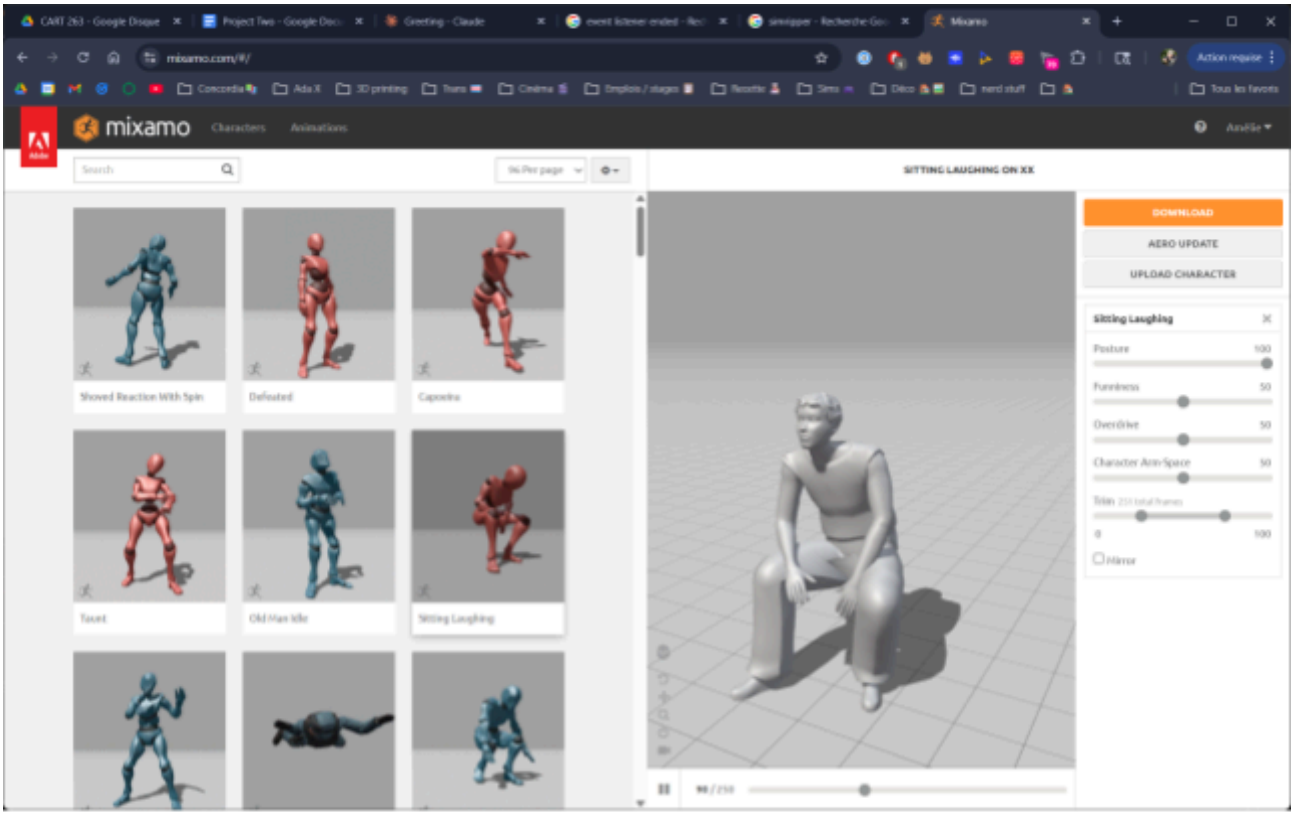
an avatar and rigging in blender



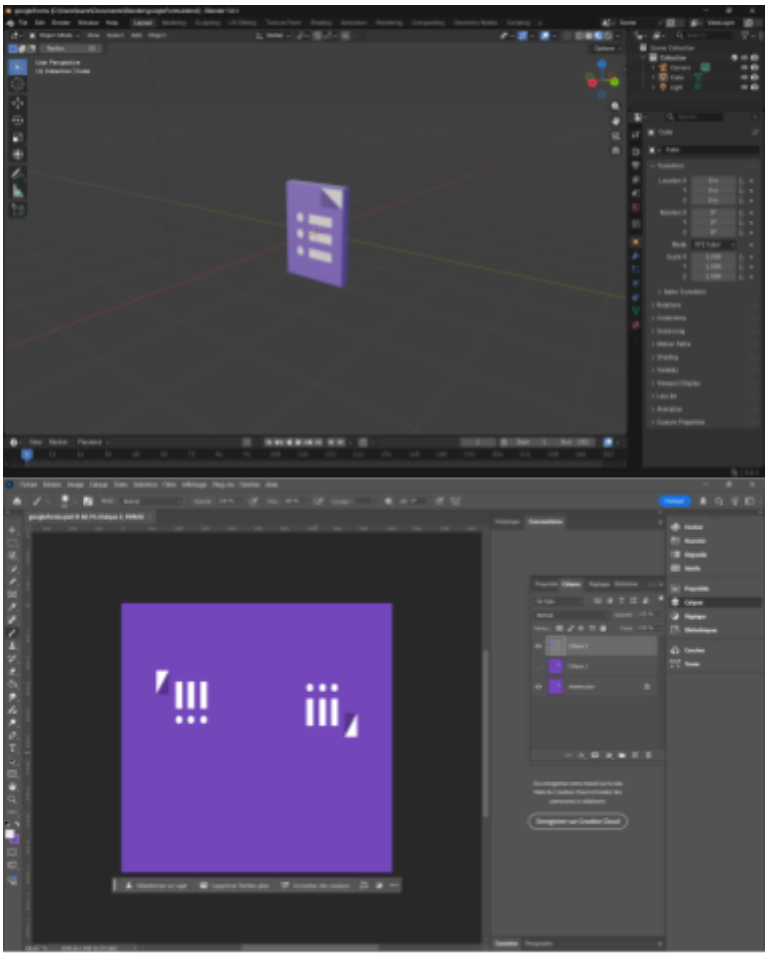
the sims studio program



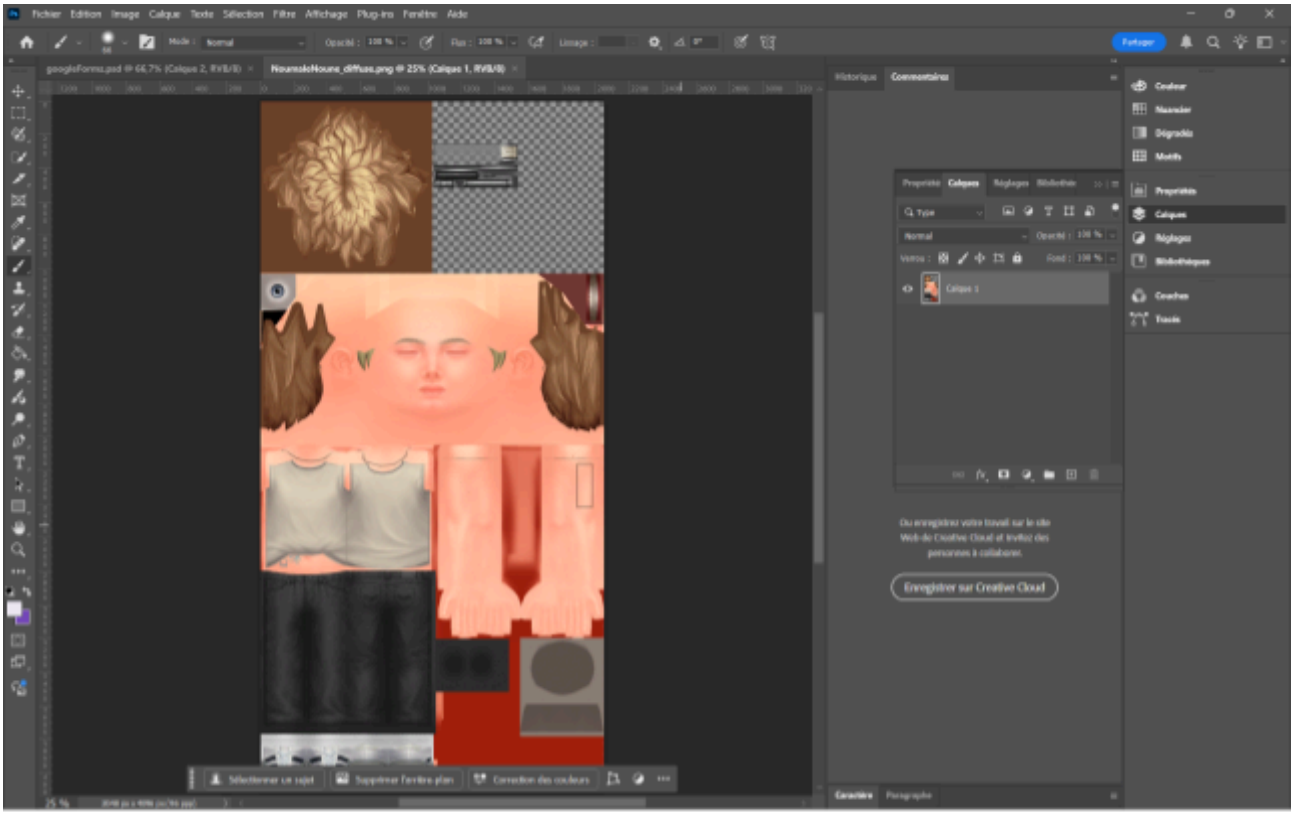
the Sims Ripper program, this isn't from my sims game as I was using a different computer



an avatar uploaded to mixamo



the google form model I created using blender and photoshop



an avatar texture in Photoshop, i changed the hair color directly on the png to match the other avatar's hair